

# Time series analysis of sap flow data to inform tree irrigation scheduling

Rebecca Killick([r.killick@lancs.ac.uk](mailto:r.killick@lancs.ac.uk))

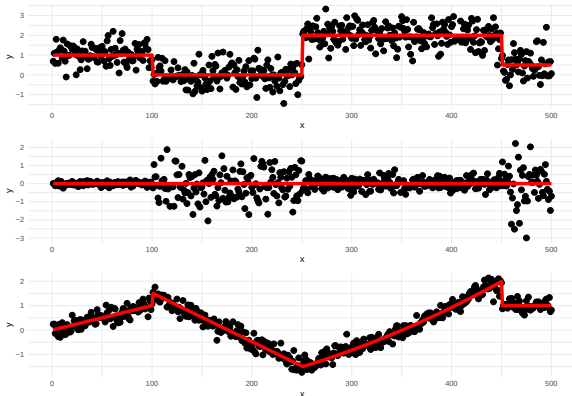
Joint work with Andy Hiron (Myerscough), Mengyi Gong & Ian Dodd (Lancaster)  
MENA-UK Workshop

- Things I'm interested in
- What is the problem?
- What is the data?
- What is the solution
  - Models
  - Pipeline

Complement Alon - focus on time rather than space

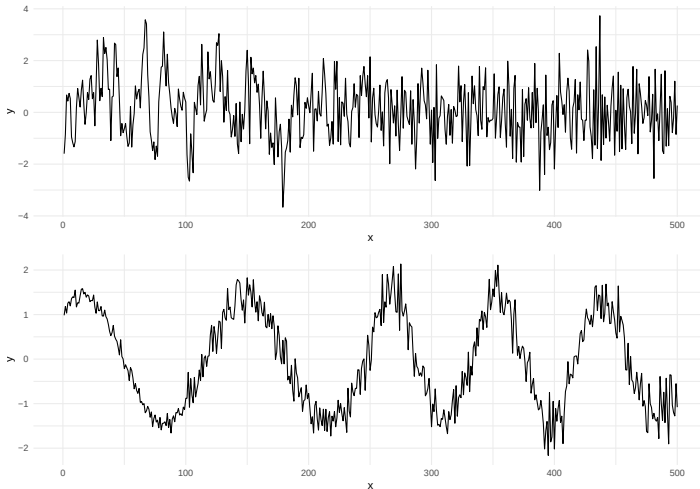
Build on Or and Zvi's talks

# What are changepoints?



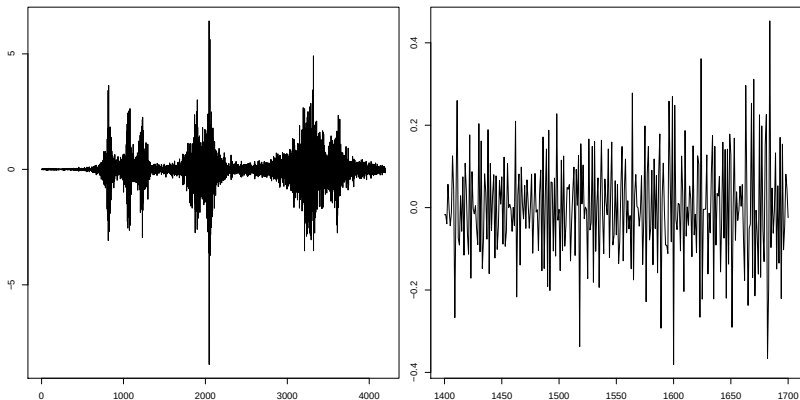
# More complicated changes

Mathematics  
& Statistics



# What is locally stationary?

Mathematics  
& Statistics



# What is the problem?

- Want to take advantage of new forms of data
  - Higher frequency (IoT)
  - Multiple sources (different frequencies)
  - Sensor behaviour not quite understood
- Not obvious what techniques to use
  - Traditional may not generalise
  - Lack of awareness of techniques
  - Lack of collaboration

# What is the problem?



We want to:

- Monitor tree health
- Understand tree water usage
- Manage tree stock; field grown and containerized

Pressures:

- Irrigation licenses
- Worry over water use (under/over)
- Need to retain class size movement



# What type of data?

## Time series!

- Sap Flow
- Dendrometer
- Environmental sensors; light, temperature, wind
- Psychrometers

## Trees (container and field):

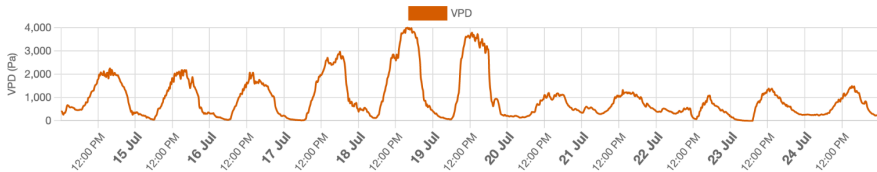
- Betula Pendula
- Carpinus Betulus
- Pyrus Calleryana



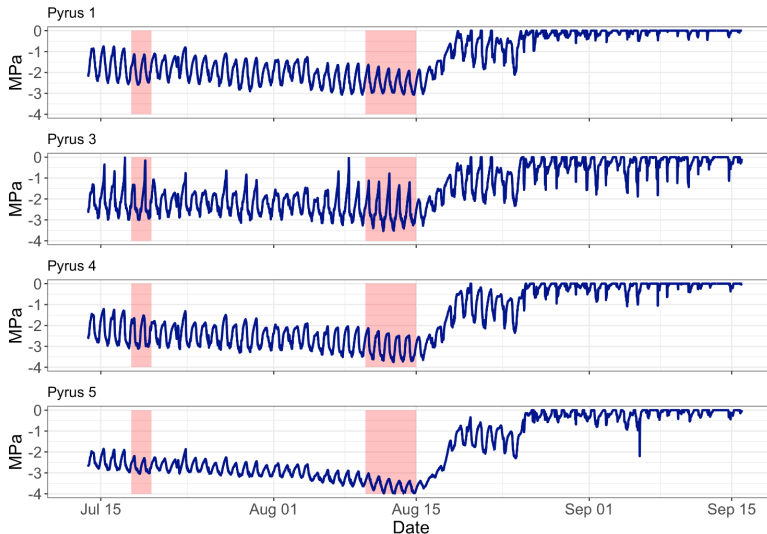
# What type of data?

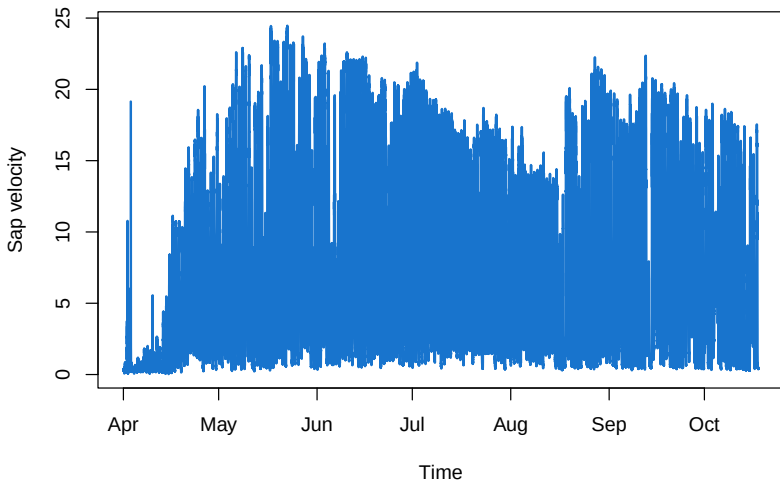
One growing season (so far):

- Over two months without rain
- Two heatwaves (July and August)
- Psychrometers down to -4 MPa (*Pyrus calleryana*)
- Vapour Pressure Deficit up to 4 kPa



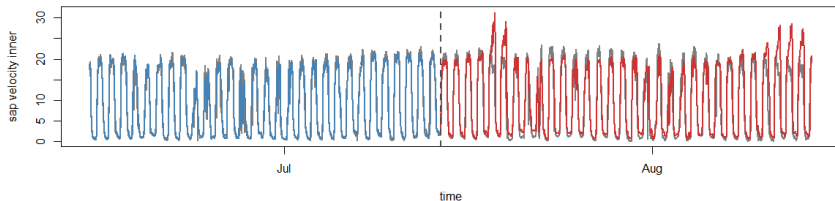
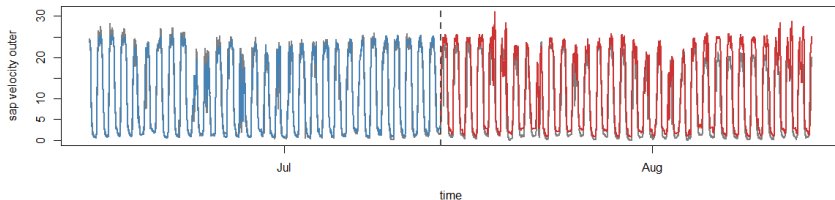
# Psychrometer Data

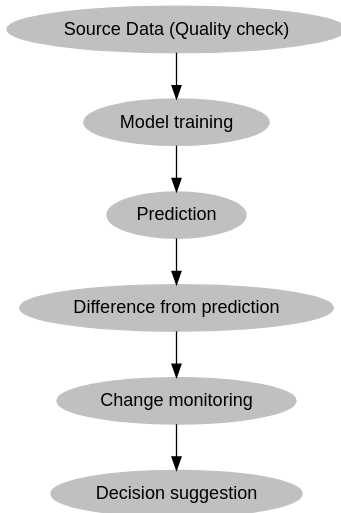




# Can we predict?

Time series modelling with weather station only.

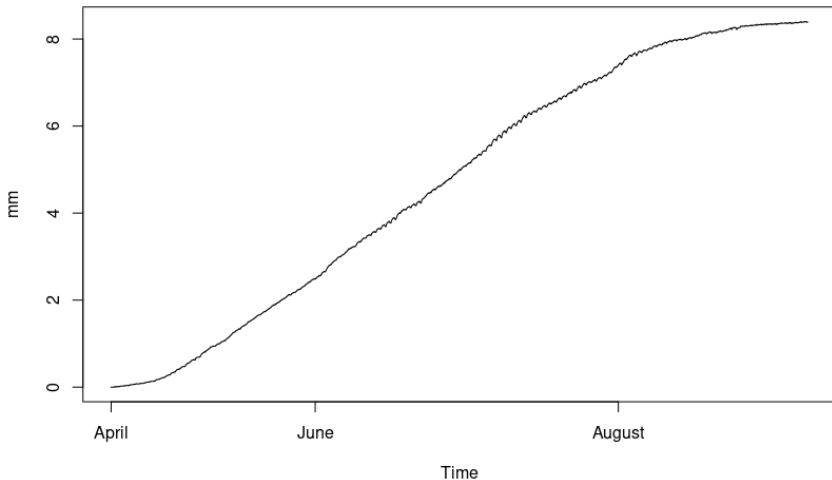




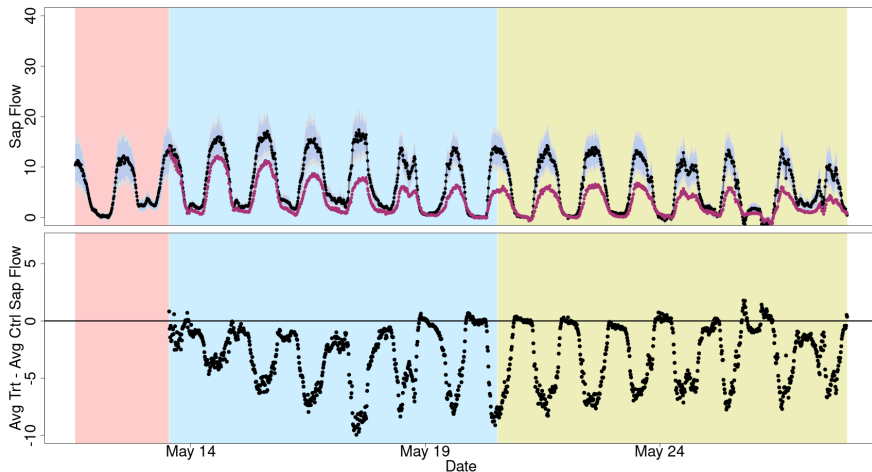
# Can we predict?



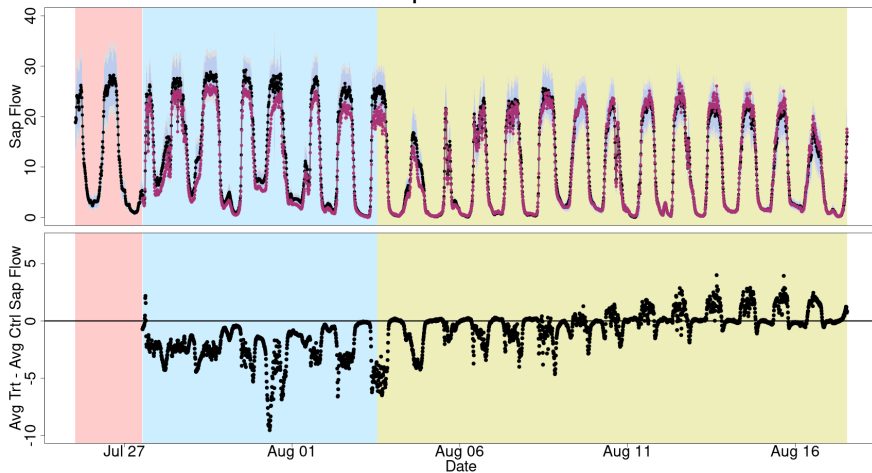
## Dendrometer



## *Prunus maackii*

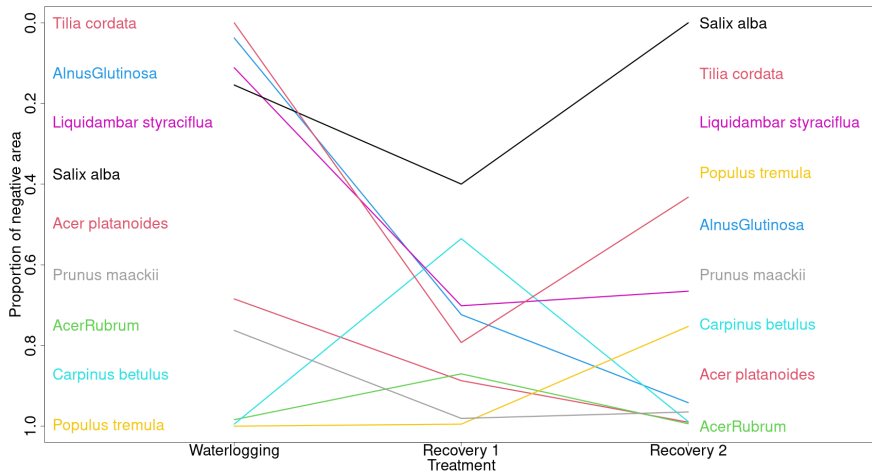


## *Populus tremula*





# Waterlogged Species



- We can use adaptive monitoring to predict behaviour
- Deviations from model used for stress monitoring
- Interaction of stress severity, length, time in season with overall growth

## Future:

- Use the learning from last season to suggest watering schedule this season
- Build trust to automate watering in 2024 season

Lots of possibilities in modern data science, be patient with us as we learn!